

AUTONOMOUS VACUUM

PROBLEM FORMULATION (DRAFT)

Group 8D

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1.0 Problems and Need

Keeping everyday living spaces clean can require a significant amount of time and physical effort. For some people, such as elderly individuals, people with limited mobility, or people with busy schedules, regular cleaning can be difficult to keep up with. Even when cleaning is manageable, it is often a repetitive task that takes time away from other activities.

1.1 Aimed Product

The aimed product is a compact autonomous vacuum-cleaning robot designed to help with everyday cleaning. The user defines the area to be cleaned and starts the robot. The robot then moves around the area on its own, detects obstacles, collects dust, debris and alerts the user when it needs attention.

Expected value: The robot should make everyday cleaning easier by reducing the amount of time and effort people need to spend vacuuming. It should also help keep the floor regularly clean without requiring constant attention from the user.

Intended users: The robot is mainly intended for people in private homes who want an easier way to keep their floors clean. It could also be useful in offices, small businesses, or other indoor spaces where regular cleaning is needed.

Operating context: The robot is intended to operate indoors on common floor surfaces such as wooden floors, tiles, and carpets. It will need to move around furniture, walls, and other objects while dealing with dust, dirt, small debris, and people or pets moving around the area.

1.2 Aimed prototype and relation to the product

The aimed prototype is a small autonomous system that demonstrates how a machine can move and operate on its own in an indoor environment. Since this is a student project, the main purpose is not to create a finished product, but to explore and understand how autonomous systems work in practice by creating a demonstrative preview of the main product. This includes investigating how the system can sense its surroundings, move around, and respond to different situations via control systems. The prototype should therefore be able to move around a defined test area, detect and avoid basic obstacles, and perform a simple cleaning-related task.

1.3 Initial Requirements

ID	Type	Requirement	Origin
R ₁	Hard	The vehicle's motion in the operating time shall not be subjected and therefore influenced by any manual input from the user	Autonomous operation
R ₂	Hard	The vehicle shall detect obstacles and avoid them	Autonomous operation, safety and longevity of the product
R ₃	Hard	The vehicle shall maintain a reasonable velocity to lower the operating time	Efficiency of the functionality
R ₄	Hard	The vehicle shall have a system to collect representative debris chosen for demonstration	Main product functionality
R ₅	Hard	The vehicle shall be able to travel to locations where cleaning could be tedious and straining for the user e.g. under a couch	Main product functionality
R ₆	Hard	The vehicle's architecture shall showcase the presence of mandatory course technologies	Mandatory course technologies
R ₇	Soft	The vehicle shall travel as equally as possible throughout the cleaning area	Cleaning range
R ₈	Soft	The debris collected should be easily discardable by the user	Accessibility of user
R ₉	Soft	The vehicle should communicate to the user that the debris collector has reached maximum capacity	Longevity of the product
R ₁₀	Soft	The vehicle should recognize when stuck and inform the user	Adjusting for possible problems
R ₁₁	Soft	The vehicle should not produce extreme noise during operation	Disturbance to user
R ₁₂	Soft	The vehicle should be easy to transport	Accessibility of user
R ₁₃	Soft	The vehicle's operation should be initialized or finalized via App by the user and physically with a button	Accessibility of user
R ₁₄	Soft	The vehicle should showcase its own battery capacity	Accessibility of user
R ₁₅	Soft	The vehicle should not scratch and damage the floor	Disturbance to user

1.4 Initial Specifications

Req.	Requirement Focus	Draft specification	Planned verification	Status
R1	Autonomous Driving	Travel autonomously for 10 minutes	Timed testing	Draft
R2	Obstacle avoidance	Completely avoid obstacles 10 x 10 cm during 5 timed tests	Obstacle avoidance test	Draft
R3	Speed criteria	The speed should be maintained at 0.2 m/s while moving	Timed distance travelled	Draft
R4	Cleaning effectiveness	The vehicle should clean a minimum of 30% of the surface. The representative debris used in the testing will be rice grains	Weight difference as percentage	Draft
R5	Traversability	The vehicle should have a maximum size of 15 cm from the floor	Measuring	Draft
R6	Course technology	At least two analogue sensors, one analogue filter, and one custom PCB.	Observation of design	Draft
R7	Area Coverage	The vehicle should cover at least 70% of the cleaning area	grid-based coverage check	Draft
R11	Noise limitation	Noise levels should not surpass 60 dB from 1 meter	dB microphone testing	Draft
R12	Transportation	The whole structure should not surpass 2.5kg of weight	Scaling test	Draft